

Label Hierarchy Transition: Delving Into Class Hierarchies to Enhance Deep Classifiers

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Abstract—Hierarchical classification aims to sort the object into a hierarchical structure of categories. For example, a bird can be categorized according to a three-level hierarchy of order, family, and species. Existing methods commonly address hierarchical classification by decoupling it into a series of multi-class classification tasks. However, such a multi-task learning strategy fails to fully exploit the correlation among various categories across different levels of the hierarchy. In this paper, we propose Label Hierarchy Transition (LHT), a unified probabilistic framework based on deep learning, to address the challenges of hierarchical classification. The LHT framework consists of a transition network and a confusion loss. The transition network focuses on explicitly learning the label hierarchy transition matrices, which has the potential to effectively encode the underlying correlations embedded within class hierarchies. The confusion loss encourages the classification network to learn correlations across different label hierarchies during training. The proposed framework can be readily adapted to any existing deep network with only minor modifications. We experiment with a series of public benchmark datasets for hierarchical classification problems, and the results demonstrate the superiority of our approach beyond current state-of-the-art methods. Furthermore, we extend our proposed LHT framework to the skin lesion diagnosis task and validate its great potential in computer-aided diagnosis.

Index Terms—Hierarchical classification, transition network, confusion loss, low-shot learning, computer-aided diagnosis.

I. INTRODUCTION

IMAGE classification is a classic computer vision task that aims to predict a category label for each image. Recent years have witnessed the tremendous success of image classification due to the rapid development of deep learning [1], especially for deep classification networks [2], [3], [4]. Conventionally, a classification network is trained with cross-entropy loss on one-hot labels, such that it predicts almost orthogonal category labels for images in different classes. Such a scheme focuses on the differences of categories, yet it may encounter challenges in effectively capturing inter-class similarity [5], [6]. Hierarchical classification, contrast to classic image classification, attempts to classify the object into a hierarchy of categories, with each hierarchy represents one specific level of concept abstraction. For example, birds can be categorized into a three-level hierarchy of order, family and species according to their taxonomic characters, as shown in Fig. 1. In hierarchical classification learning, the labels can be organized according to the domain knowledge (e.g., WordNet database [7]) or automatically generated by algorithms [8], [9], [10], [11], [12].

Compared to traditional multi-class classification, hierarchical classification pays more attention to exploit rich semantic correlation among categories [13], in order to improve the performance of traditional classification tasks [14], [15] or mitigate the severity of prediction mistakes [16], [17], i.e., making the incorrectly classified samples fall within semantically related categories. Take the example of making diagnosis for one malignant tumor, mistaking it as another malignant one in the same subtype should be much less serious than mistaking it as one benign tumor, as such a mistake would have crucial implications in terms of the follow-up diagnosis and treatment.

Hierarchical classification has been proved effective in many visual scenarios, which can be traced back to traditional linear classifiers [19], [20], [21], [22]. Recently, hierarchical classification has attracted much attention in deep learning [23]. Most of the existing works [16], [24], [25], [26], [27] commonly fall in a multi-task learning manner, which employs a hierarchy-agnostic feature extractor to learn universal representations across hierarchies and parameterizes a set of hierarchy-specific heads to encode private representations for each hierarchy. For example, a deep network was proposed in [24] to classify coarse- and fine-level categories through two different branches. In [16],

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The code of our method is available at <https://github.com/renzhenwang/label-hierarchy-transition>.

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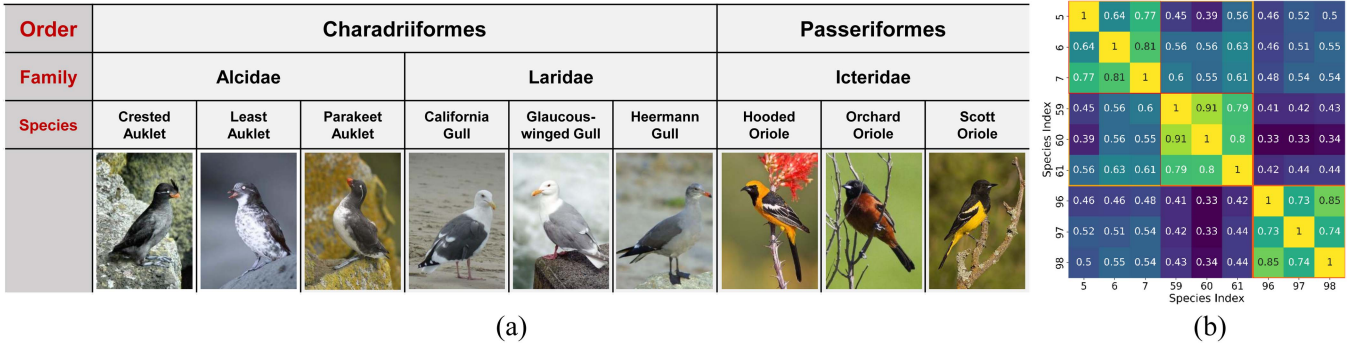


Fig. 1. Hierarchical structure in CUB-200-2011 [18]. (a) Example taxonomy (*Species–Family–Order*) with representative bird images, where visual similarity is preserved within hierarchies. (b) Pairwise similarity matrix computed from 312 attributes, showing high correlations within hierarchies and obvious distinctions across coarser-level ones.

a granular network was proposed for food image classification, where a single network backbone was shared by multiple fully-connected layers with each one responsible for the label prediction within one hierarchy. Chen et al. [26] likewise employed a multi-head architecture to output hierarchy-specific prediction, but instead introduced an attention mechanism to guide label prediction of fine-level hierarchies through that of the coarse-level hierarchies. Furthermore, Alsallakh et al. [25] used a standard deep architecture to fit the finest-level labels, and added branches at the intermediate layers to fit the coarser-level labels. Recently, Chang et al. [27] pointed out that fine-level features benefit coarse-grained classifier learning, yet coarse-level label prediction is harmful to fine-level feature learning. They in turn proposed to disentangle coarse- and fine-level features by leveraging hierarchy-specific classification heads, and further force finer-level features (with stopped gradient) to participate in coarser-level label predictions. Such a simple method has achieved significant improvement in fine-level image recognition.

Despite significant progress, these recent methods predict category labels for each hierarchy using independent classification heads. This implies that all class hierarchies are statistically independent (to be elaborated in Section III), such that category correlations across hierarchies are encoded solely by a shared feature extractor. Such a multi-task learning strategy is often sub-optimal, as it may lead to the common issue of under-transfer as knowledge can not be sufficiently transferred across different classification heads [28]. Moreover, these recent methods primarily focus on a restrictive setting where labels for all hierarchies are provided during the training phase. In contrast, the more challenging yet realistic setting, where samples are labeled at only partial hierarchies, remains under-explored [29].

To address the aforementioned issues, this paper aims to develop a deep hierarchical classification method capable of *explicitly* capturing category correlations embedded in class hierarchies. Unlike prior work that *implicitly* encodes category correlations through a universal feature extractor, our approach explicitly connects category predictions between adjacent hierarchies. We achieve this by using transition probabilities of category labels from one hierarchy to the next, based on the input image or features. Concretely, we propose Label Hierarchy

Transition (LHT), a unified probabilistic approach, to recursively predict the labels from fine- to coarse-level class hierarchies. The LHT model contains two components: (1) an arbitrary existing classification network, which is used to predict the labels for the finest-level hierarchy; (2) a transition network, which generates a label hierarchy transition matrix for each hierarchy (except for the finest-level one). Each element of the matrix denotes the transition probability of one class at the corresponding hierarchy given the other class of the adjacent finer-level hierarchy. For each coarser-level hierarchy, the category prediction can be formulated as the product of its corresponding transition matrix and the prediction score of its adjacent finer-level hierarchy. The two components are jointly trained in an end-to-end manner, and the learned transition matrices for coarser-level hierarchies theoretically provide supervision for the finer-level ones. This enables our proposed LHT framework significantly improve the performance of hierarchical classification, especially in scenarios where a large number of samples have labels only at partial hierarchies.

Moreover, the hierarchical cross-entropy loss used in prior works based on multi-task learning tends to produce orthogonal predictions at each hierarchy and is not conducive to the learning of correlations among different categories. To address this challenge, we further propose a confusion loss to explicitly regularize the label hierarchy transition matrices, which facilitates the network to learn category correlations from all class hierarchies during the training phase.

In summary, the main contributions are four-fold:

- We propose a unified deep classification framework to address the hierarchical classification problem. Contrast to predicting the labels for each hierarchy, we propose to learn label hierarchy transition matrices, which encode the likelihood of similarity between classes across different label hierarchies. To the best of our knowledge, we are the first to utilize deep networks to explicitly learn transition probabilities across class hierarchies.
- We provide grounded theoretical understanding that explains how our method potentially works in real-world scenarios. Specifically, we prove that the proposed method facilitates bi-directional information flow between fine- and coarse-level hierarchies, enabling both granular detail

enhancement and broader contextual integration. Furthermore, the learned transition matrices at coarse-level hierarchies serve as supervisory signals for fine-level ones, offering a practical advantage in missing-label scenarios.

- We propose a confusion loss to directly regularize the negative entropy of the column space of label hierarchy transition matrices, which further encourages the network to learn correlations among class hierarchies, and alleviates the issue of over-confident predictions arising from cross-entropy loss.
- Our LHT method can be readily applied to any existing classification network, and the resulting model can be trained in an end-to-end manner. Extensive experiments on a series of hierarchical classification tasks demonstrate the superiority of our approach beyond the prior state-of-the-arts.

The rest of this paper is organized as follows. Section II revisits related works. Section III describes the formulation of the proposed LHT framework as well as its training algorithm. Section IV presents experimental results and analysis in detail. Section V provides a realistic application of our method in computer-aided diagnosis. Section VI summarizes the final conclusion.

II. RELATED WORK

Hierarchical classification [30], [31], [32], exploiting a hierarchical structure of labels for classification task, has been extensively explored in natural language processing [33], [34], [35], computer vision and computer-aided diagnosis [36], [37], [38]. In the existing literature, hierarchical classification is mainly related to three aspects of research: hierarchical architecture, hierarchical loss function, and label embedding based methods. Next, we discuss related studies with respect to these three aspects.

A. Hierarchical Architecture

A common way of designing a hierarchical architecture is to alleviate the computational burden of traditional linear classifiers when the number of recognition categories become huge [22], [39], [40], [41], [42]. The core idea was to create a tree-like classifier whose leaf nodes were used for the prediction of the given categories and internal nodes for the cluster of categories.

Most recently, deep learning based methods are increasingly used to address hierarchical classification [16], [24], [25], [26], [27], [29], [43]. These methods commonly fall into a multi-task learning manner, and adopt shared-specific feature representation learning to fit labels across different levels of hierarchy. Such a flexible architecture is advantageous in learning low-level features on one hand, while on the other hand, it can easily incorporate canonical inclusion relations between the parent node (super-class) and its child nodes (sub-classes) [27], [43] or exclusion relations [29] between the sibling nodes at the same hierarchy into the network design. Contrary to these works, our proposed LHT framework learns the conditional probabilities

between two arbitrary nodes within adjacent hierarchies to encode the affinity relations.

B. Hierarchical Loss Function

A range of literature on hierarchical classification resorts to incorporating the class hierarchies in the loss function. In [44], the metrics associated with each node of the taxonomy tree were learned based on a probabilistic nearest-neighbour classification framework. In [45], the tree-structure hierarchies were used to impose prior on the parameters of the classification layer for knowledge transfer between classes, and similar strategy was proposed by Zhao et al. [21] in the traditional linear classifier. In [23], various label relations (e.g., mutual exclusion and subsumption) were encoded as conditional random field to pair-wisely capture the correlation between classes. In [46], max constraint loss was proposed to produce predictions coherent with the hierarchy constraint for multi-label hierarchical classification. Recently, Bertinetto et al. [17] proposed a hierarchical loss that could be viewed as a weighted cross-entropy loss, and the weights were defined according to the depth of the label tree. In parallel to our method, chen et al. [29] proposed a probabilistic loss to encode the label relation within each hierarchy. Unlike these methods [47], [48], we combine cross-entropy with a novel confusion loss to regularize the learned transition matrices, thereby fully exploiting class correlations across hierarchies. Besides, in hierarchical segmentation [47], [48], [49], prior works such as HSSN [47], [48] incorporates class hierarchies by directly constraining the output space through a tree-min loss to produce hierarchy-coherent predictions. In contrast, our LHT takes a probabilistic approach by learning transition matrices to model inter-class similarities, which not only enforces hierarchy coherence, but also leverages cross-level correlations more flexibly, especially under missing-label scenarios.

C. Label Embedding

Label embedding aims to encode the correlation information among classes into the traditional one-hot labels, which has been widely explored in multi-class classification [5], [6], [50], [51]. For example, Label Smoothing presented in [5], [6] softens the one-hot label by assigning a slightly lower probability to the target class and distributes the remaining probability evenly across the non-target classes. In hierarchical classification, label embedding techniques typically map one-hot labels to vectors, where the relative positions of these vectors encode the structural information embedded in the class hierarchy. Specifically, Bertinetto et al. [17] proposed Soft Labels, an embedding-based method that softens one-hot labels using a metric derived from the lowest common ancestor (LCA) height within a given label tree. Dhall et al. [52] employed entailment cones to learn order-preserving label embeddings and subsequently integrated them with image embeddings for hierarchical classification by using the transitivity of the embeddings across levels in the hierarchy. Although label embedding is a potential strategy to capture the correlation across class hierarchies, in this paper, we attempt to develop a unified deep classification framework for explicitly encoding correlation of hierarchical classification.

III. THE PROPOSED METHOD

In this section, we first introduce the problem formulation of hierarchical classification and the preliminaries of our work. We then provide an overview of our proposed LHT framework that explicitly encodes category correlations across distinct hierarchies. Next, we present the architecture of the proposed transition network. Finally, we introduce the proposed confusion loss to encourage correlation learning across hierarchies.

A. Problem Formulation and Preliminaries

In hierarchical classification tasks, a given image $\mathbf{x}$ is typically annotated with a sequence of K labels $(y^1, y^2, \dots, y^K)$, where K represents the number of label hierarchies. Each $y^k \in \{1, 2, \dots, C_k\}$ is the label within the k -th hierarchy, with C_k denoting the number of categories of this hierarchy. We denote $\mathbf{y}^k$ is the corresponding one-hot vector of y^k . Without loss of generality, we define $k = 1$ as the finest-level hierarchy, and assume the granularity becomes coarser as k increases. Accordingly, the number of classes decreases with larger k , i.e., $C_k > C_{k+1}$. The goal of hierarchical classification is to learn a classifier to predict hierarchical labels for each unknown $\mathbf{x}$.

The multi-task approach to addressing hierarchical classification commonly involves two steps: i) it first employs a hierarchy-agnostic feature extractor f_ω parameterized by ω to learn a universal representation $\hat{\mathbf{x}} = f_\omega(\mathbf{x})$ across hierarchies, and ii) for the k -th hierarchy, it uses a hierarchy-specific classification head $g_{\theta^k}^k$ parameterized by θ^k to predict the corresponding classification distribution $\mathbf{p}^k = g_{\theta^k}^k(\hat{\mathbf{x}})$. Note that here, with some abuse of notation, we define $\mathbf{p}^k = [p(Y^k = 1|\mathbf{x}), \dots, p(Y^k = C_k|\mathbf{x})]^T$, where Y^k denotes a random variable representing the label of the k -th hierarchy. The involved parameters are jointly optimized using a hierarchical cross-entropy loss

$$\mathcal{L}_{\text{CE}} = \sum_{n=1}^N \sum_{k=1}^K H(\mathbf{p}_n^k, \mathbf{y}_n^k), \quad (1)$$

where $\{\mathbf{x}_n, (\mathbf{y}_n^1, \dots, \mathbf{y}_n^K)\}_{n=1}^N$ represents a batch of N training samples, and $H(\cdot, \cdot)$ is the cross-entropy loss defined as $H(\mathbf{p}, \mathbf{y}) = \sum_i \mathbf{y}[i] \log \mathbf{p}[i]$, with $\mathbf{p}[i]$ denoting the i -th entry of $\mathbf{p}$.

Eq. (1) is equivalent to solve the negative log-likelihood for $p(y^1, \dots, y^K|\mathcal{I}) = \prod_{k=1}^K p(y^k|\mathcal{I})$. The detailed derivation can be found in Appendix A.1, available online. This implies that all class hierarchies are considered statistically independent of each other, such that category correlations across hierarchies are encoded solely by the shared feature extractor $f_\omega(\cdot)$. Such a multi-task learning strategy is often sub-optimal due to the following two reasons: i) It fails to explicitly model the hierarchy/task relationships, which may lead to the common issue of under-transfer as knowledge can hardly be transferred across different classification heads [28]. ii) In missing-label scenarios where most samples are labeled at only partial hierarchies, there is a risk that the classification head associated with the finest-level hierarchy might overfit, as it is trained on fewer samples compared to the coarser-level hierarchies.

In light of the aforementioned reasons, a natural question is how to explicitly model hierarchical relationships within a multi-task deep learning framework to facilitate information transfer across different classification heads, especially in challenging scenarios with missing labels.

B. Label Hierarchy Transition

The goal of this paper is to propose a unified probabilistic framework to explicitly encode category correlations across hierarchies into the existing multi-task learning framework. Unlike previous works based on the assumption of independent distributions across different hierarchies, we mathematically consider that the label Y^k of the k -th hierarchy is statistically dependent on Y^{k-1} and the sample $\mathbf{x}$ itself. It is natural to encode such correlations between adjacent hierarchies through the conditional distribution $p(Y^k|Y^{k-1}, \mathbf{x})$. Subsequently, we can formulate the distribution $p(Y^k|\mathbf{x})$ by marginalizing over the variable Y^{k-1} :

$$p(Y^k|\mathbf{x}) = \sum_{i=1}^{C_{k-1}} p(Y^k|Y^{k-1} = i, \mathbf{x}) \cdot p(Y^{k-1} = i|\mathbf{x}), \quad (2)$$

where $p(Y^{k-1}|\mathbf{x})$ is the label distribution of the $(k-1)$ -th hierarchy with $k = 2, \dots, K$.

For simplicity, here we define

$$t_{ij}^k \triangleq p(Y^k = i|Y^{k-1} = j, \mathbf{x}), \quad (3)$$

and then we have a $C_k \times C_{k-1}$ matrix

$$\mathbf{T}^k = \begin{bmatrix} t_{11}^k & t_{12}^k & \dots & t_{1C_{k-1}}^k \\ t_{21}^k & t_{22}^k & \dots & t_{2C_{k-1}}^k \\ \vdots & \vdots & \ddots & \vdots \\ t_{C_k1}^k & t_{C_k2}^k & \dots & t_{C_kC_{k-1}}^k \end{bmatrix}. \quad (4)$$

We call this matrix *Label Hierarchy Transition Matrix* (or abbreviated as *Transition Matrix*), since it defines the probabilities of transitions from fine-grained labels to coarse-grained labels and we can get a glimpse of the *transition* status between the adjacent class hierarchies. Note that the elements of $\mathbf{T}^k$ are positive and each column sums to 1, i.e., $\sum_{i=1}^{C_k} t_{ij}^k = 1$ for each $j \in \{1, \dots, C_{k-1}\}$.

1) *Model Formulation*: In this paper, the core of our *Label Hierarchy Transition* model is to predict the label distribution $p(Y^k|\mathbf{x})$ for each hierarchy, by jointly learning the two kinds of conditional probability involved in (2). Notably, (2) inherently defines a fine-to-coarse learning paradigm, as illustrated in Fig. 2. Within this paradigm, the finest-level label distribution is first fitted, and the coarser-level ones are subsequently learned, conditioned on the input image and its adjacent finer-level label distribution. Specifically:

- For the finest-level hierarchy, we use a vanilla *Classification Network* f_ω with parameters ω to predict its label distribution $\mathbf{p}^1$ as follows:

$$\mathbf{p}^1 = \sigma(\mathbf{z}) = \sigma(f_\omega(\mathbf{x})), \quad (5)$$

where $\mathbf{p}^1 = [p(Y^1 = 1|\mathbf{x}), \dots, p(Y^1 = C_1|\mathbf{x})]^T$, $\mathbf{z}$ denotes the logits of $\mathbf{x}$, and σ is the softmax function.

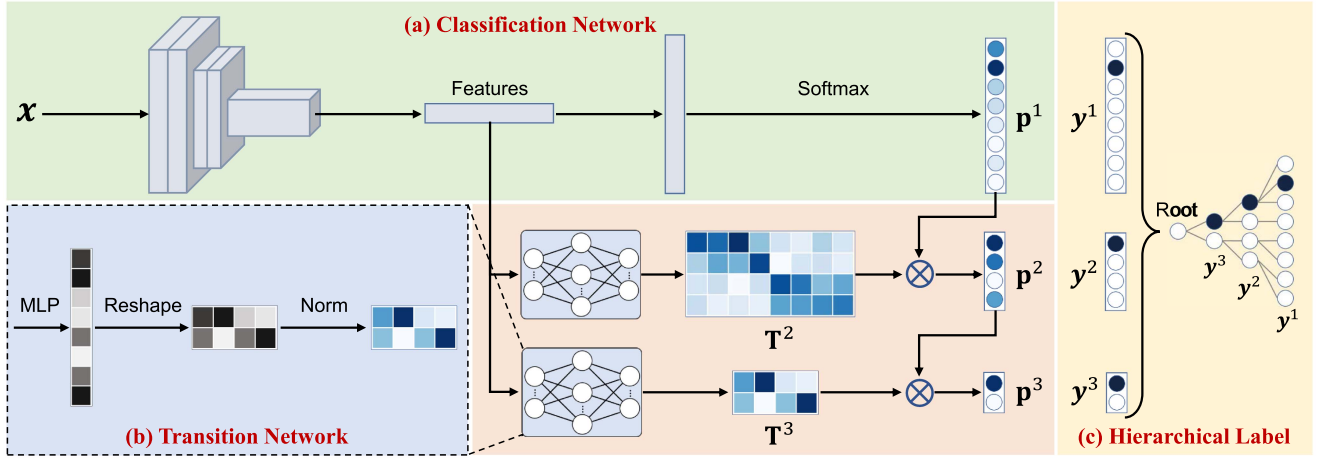


Fig. 2. Schematic illustration of the proposed LHT model. We take a three-level hierarchical classification as an example, where the number of classes within each hierarchy is 8, 4 and 2, respectively. The model consists of two key components: (i) Classification Network, which can be any existing classification network for predicting the labels of the finest-level hierarchy; (ii) Transition Network, which leverages a light-weight network (e.g., a multi-layer perceptron (MLP) with one single hidden layer) to map the feature embedding of the input sample to the label hierarchy transition matrix for each hierarchy. For each coarse-level hierarchy, its label prediction is obtained by multiplying the corresponding transition matrix by the label prediction of its adjacent fine-level hierarchy.

- For the k -th level hierarchy with $k = 2, \dots, K$, we design a light-weight *Transition Network* $g_{\theta^k}^k$ parameterized by θ^k to predict its transition matrix $\mathbf{T}^k$, such that $\mathbf{T}^k = g_{\theta^k}^k(\mathbf{x})$. Thus, the label distribution $\mathbf{p}^k$ can be formulated as

$$\mathbf{p}^k = \mathbf{T}^k \mathbf{p}^{k-1}. \quad (6)$$

The architectural designs of $g_{\theta^k}^k$ will be elaborated in the next subsection.

From (5) and (6), we can get the label distributions of all hierarchies. This allows for the joint optimization of the parameters involved in both the Classification Network and the Transition Networks, i.e., $\{\omega, \theta^2, \dots, \theta^K\}$, in an end-to-end manner. Given a batch of N training samples $\{\mathbf{x}_n, (\mathbf{y}_n^1, \mathbf{y}_n^2, \dots, \mathbf{y}_n^K)\}_{n=1}^N$, the hierarchical cross-entropy loss can be formulated as follows:

$$\begin{aligned} \mathcal{L}_{\text{CE}} &= \sum_{n=1}^N \sum_{k=1}^K H(\mathbf{p}_n^k, \mathbf{y}_n^k) \\ &= \sum_{n=1}^N \left(H(\mathbf{p}_n^1, \mathbf{y}_n^1) + \sum_{k=2}^K H(\mathbf{T}_n^k \mathbf{p}_n^{k-1}, \mathbf{y}_n^k) \right). \end{aligned} \quad (7)$$

In addition to (7), we propose a novel confusion loss to enhance the learning of category correlations across all class hierarchies during the training phase. The details will be discussed in Section III-D.

2) *Theoretical Analysis*: Our LHT framework defined by (5) and (6) offers the following theoretical guarantees: i) LHT facilitates the transition of information from the finest-level hierarchy to the coarser-level ones via forward propagation; ii) LHT enables the transition of true label information from coarser-level hierarchies to the finer-level ones through back-propagation; iii) The learned transition matrices for coarser-level hierarchies theoretically provide supervised signals for the finer-level hierarchies, thereby benefiting the training process for missing-label scenarios.

For ease of notation, we only consider the single-sample case ($N = 1$) for (7), but the derivation for the multi-sample case is straightforward. We first introduce the definition of the transition matrix from the 1-st hierarchy to the k -th hierarchy.

Definition 1: We define the following matrix

$$\mathbf{T}^{k \leftarrow 1} = \begin{bmatrix} t_{11}^{k \leftarrow 1} & t_{12}^{k \leftarrow 1} & \dots & t_{1C_1}^{k \leftarrow 1} \\ t_{21}^{k \leftarrow 1} & t_{22}^{k \leftarrow 1} & \dots & t_{2C_1}^{k \leftarrow 1} \\ \vdots & \vdots & \ddots & \vdots \\ t_{C_k 1}^{k \leftarrow 1} & t_{C_k 2}^{k \leftarrow 1} & \dots & t_{C_k C_1}^{k \leftarrow 1} \end{bmatrix} \quad (8)$$

as the **transition matrix from the 1-st hierarchy to the k -th hierarchy**, where $t_{ij}^{k \leftarrow 1} = p(Y^k = i | Y^1 = j, \mathbf{x})$ is the transition probability from the j -th class within the 1-st hierarchy to the i -th class within the k -th hierarchy.

Given the definition of $\mathbf{T}^{k \leftarrow 1}$, we show that our proposed LHT facilitates the transition of information from the finest-level hierarchy directly to any coarser-level ones as in the following Proposition 1.

Proposition 1: In the LHT framework, the label prediction of the k -th ($k > 1$) hierarchy can be inferred from the finest-level hierarchy as follows:

$$\mathbf{p}^k = \mathbf{T}^{k \leftarrow 1} \mathbf{p}^1, \quad (9)$$

where the transition matrix $\mathbf{T}^{k \leftarrow 1}$ is defined as $\mathbf{T}^{k \leftarrow 1} := \mathbf{T}^k \mathbf{T}^{k-1} \dots \mathbf{T}^2$, which is implemented by the composite function $g_{\theta^k}^k \circ g_{\theta^{k-1}}^{k-1} \circ \dots \circ g_{\theta^2}^2$.

The proposition can be proved by verifying that each (i, j) -th element of $\mathbf{T}^{k \leftarrow 1}$ represents the transition probability $p(Y^k = i | Y^1 = j, \mathbf{x})$. We provide the proof in Appendix A.2, available online. Proposition 1 indicates that the label prediction for the k -th hierarchy is conditioned on the information derived from all the preceding $k - 1$ hierarchies. This, on the one hand, demonstrates that our LHT framework facilitates the transition of information from fine-level hierarchies to coarse-level ones

through the forward propagation of the framework. On the other hand, since LHT optimizes all parameters end-to-end via (7), it implies that LHT also benefits the transition of true label information from coarser-level hierarchies to the finest-level one through gradient backpropagation, as illustrated in the following proposition.

Proposition 2: If the loss function in (7) are minimized by gradient based algorithms (such as SGD), then the loss term $\mathcal{L}^k \triangleq H(\mathbf{p}^k, \mathbf{y}^k)$ enables the propagation of true label information from coarse-level hierarchies to the finest-level one through gradient backpropagation. Specifically, the gradient of $\mathcal{L}^k$ with respect to the logits $\mathbf{z}$ of the finest-level hierarchy is computed as

$$\mathcal{G}_j^k := \frac{\partial \mathcal{L}^k}{\partial \mathbf{z}[j]} = \left(1 - \frac{t_{y^k j}^{k \leftarrow 1}}{\mathbf{p}^k[y^k]}\right) \mathbf{p}^1[j], \quad (10)$$

where $\mathbf{z}[j]$ denotes the j -th entry of $\mathbf{z}$ and $k = 2, \dots, K$.

The proof of Proposition 2 is provided in Appendix A.3, available online. Eq. (10) demonstrates that the coarse-level hierarchies influence the finest-level hierarchy by adjusting its logit $\mathbf{z}$ using the true label y_k from the k -th hierarchy in the end-to-end training phase. Intuitively, if $\mathcal{G}_j^k < 0$, reducing the loss $\mathcal{L}^k$ by updating parameters along the negative gradient direction ($-\mathcal{G}_j^k$) will effectively increase $\mathbf{z}[j]$; conversely, if $\mathcal{G}_j^k > 0$ the same process will lead to a decrease in $\mathbf{z}[j]$. In fact, the Transition Networks can rectify inaccurate predictions by adjusting the logits of the finest-level hierarchy during training. This can be formalized in the following corollary.

Corollary 1: If the network parameters are updated according to (7) using SGD-based optimizers, then $\mathcal{G}_{j^*}^k < 0$ holds for $j^* = \arg \max_j t_{y^k j}^{k \leftarrow 1}$, which is the subclass most visually similar to y^k as determined by the Transition Networks.

This corollary indicates that reducing $\mathcal{L}^k$ along the direction ($-\mathcal{G}_{j^*}^k$) can guide the logits $\mathbf{z}$ of the finest-level hierarchy towards the value that yield higher probability for the subclass most visually similar to y^k during training.

Intrinsically, in Proposition 2, $\mathcal{G}_j^k$ measures the distance between two distributions. By the definition of transition probability in (3), $\mathcal{G}_j^k$ can be rewritten as:

$$\begin{aligned} \mathcal{G}_j^k &= \left(1 - \frac{t_{y^k j}^{k \leftarrow 1}}{\mathbf{p}^k[y^k]}\right) \mathbf{p}^1[j] \\ &= p(Y^1 = j | \mathbf{x}) - \frac{p(Y^k = y^k | Y^1 = j, \mathbf{x}) p(Y^1 = j | \mathbf{x})}{p(Y^k = y^k)} \\ &= p(Y^1 = j | \mathbf{x}) - p(Y^1 = j | Y^k = y^k, \mathbf{x}), \end{aligned} \quad (11)$$

where the third equal sign holds due to Bayes' rule. Eqs. (10) and (11) further demonstrate that the k -th hierarchy can facilitate logit adjustment for the finest-level one by modifying the distance between $p(Y^1 | \mathbf{x})$ and $p(Y^1 | Y^k = y^k, \mathbf{x})$ during the end-to-end training phase. Especially in missing-label scenarios, (10) and (11) indicate that the coarser-level hierarchies can provide supervision for the finest-level hierarchy. In fact, the adjustment induced by the k -th hierarchy positively impacts the label prediction of the finest-level hierarchy. This can be summarized in the following theorem:

Theorem 1: In missing-label scenarios where the training sample $\mathbf{x}$ is partially annotated at the k -th hierarchy with ground-truth label y^k . If all entries of the feature vector input to the linear classification layer are non-zero,¹ and the network parameters are updated by $\mathcal{L}^k \triangleq H(\mathbf{p}^k, \mathbf{y}^k)$ using SGD-based optimizers, then at the minimum, the following equality holds:

$$p(Y^1 | \mathbf{x}; \omega) = p(Y^1 | Y^k = y^k, \mathbf{x}; \omega, \theta), \quad (12)$$

where $p(Y^1 | \mathbf{x}; \omega)$ is formulated by the Classification Network with parameters ω , and $p(Y^1 | Y^k = y^k, \mathbf{x}; \omega, \theta)$ is jointly formulated by the Classification Network and the Transition Networks with parameters ω and $\theta = \{\theta^1, \dots, \theta^k\}$, respectively.

We present the proof of Theorem 1 in Appendix A.5, available online. Since $p(Y^1 | Y^k = y^k, \mathbf{x}; \omega, \theta)$ represents the predicted label distribution at the finest-level hierarchy given that the k -th hierarchy's prediction is the ground-truth label, this theorem implies that our LHT framework aims to encourage the prediction of the Classification Network $p(Y^1 | \mathbf{x}; \omega)$ to better match the conditional distribution $p(Y^1 | Y^k = y^k, \mathbf{x}; \omega, \theta)$. In summary, it concludes that:

- The coarse-level hierarchies can provide supervision for the finest-level hierarchy by implicitly minimizing the distance between the two probability distributions $p(Y^1 | \mathbf{x}; \omega)$ and $p(Y^1 | Y^k = y^k, \mathbf{x}; \omega, \theta)$.
- The Classification Network tends to predict the labels for the finest-level hierarchy while maintaining the accuracy of the k -th hierarchy's prediction during the end-to-end training phase.

C. Architectural Designs

As aforementioned, the proposed LHT framework mainly consists of the Classification Network and the Transition Networks.

Firstly, we emphasize that for the Classification Network as formulated in (5), we employ a well-designed existing network, such as ResNet-50 [4], to predict category labels for the finest-level hierarchy. On one hand, this aims to insure a fair comparison of classification performance at the finest-level hierarchy with traditional hierarchical classification methods based on the multi-task learning paradigm, while on the other hand, the target is to demonstrate that our proposed LHT is a plug-in for enhancing conventional classification networks through hierarchical supervision, especially in missing-label scenarios.

Next, we elaborate the architectural designs for the Transition Networks. As shown in Fig. 2(b), for hierarchy k , the network $g_{\theta^k}^k$ is responsible for producing the transition matrix $\mathbf{T}^k$. A natural way to implement the network is to use a single fully-connected (FC) layer, which can be formulated as:

$$\mathbf{T}^k = g_{\theta^k}^k(\mathbf{x}) := \text{softmax}(\text{reshape}(\mathbf{W}^k \hat{\mathbf{x}} + \mathbf{b}^k)), \quad (13)$$

where $\hat{\mathbf{x}} \in \mathbb{R}^d$ is a high-level embedding of $\mathbf{x}$ branched from the Classification Network, $\theta^k = \{\mathbf{W}^k, \mathbf{b}^k\}$ are the parameters of

¹This assumption is practical, as we can introduce a small positive bias term (e.g. $\epsilon = 10^{-8}$) or directly apply a sigmoid activation before the linear classification layer to ensure the input is greater than 0.

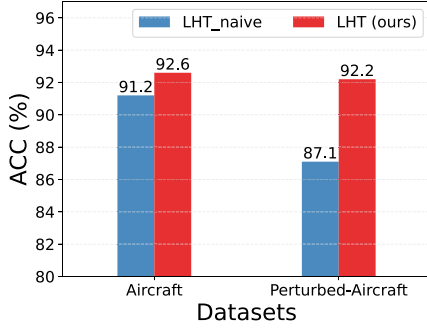


Fig. 3. Performance comparison between LHT_naive and LHT (ours) on the Aircraft and Perturbed-Aircraft datasets.

$g_{\theta^k}^k$, with $\mathbf{W}^k \in \mathbb{R}^{(C_k \times C_{k-1}) \times d}$ and $\mathbf{b}^k \in \mathbb{R}^{(C_k \times C_{k-1})}$ denoting the weight matrix and bias term of the FC layer, respectively. The operator $\text{reshape}(\cdot)$ rearranges the elements of a $(C_k \times C_{k-1})$ vector into a $C_k \times C_{k-1}$ matrix, and $\text{softmax}(\cdot)$ normalizes the columns of the input matrix using the softmax function for satisfying $\sum_{i=1}^{C_k} \mathbf{T}_{ij}^k = 1$.

From (13), we can observe that the Transition Networks capture sample-specific information through the weight matrix $\mathbf{W}^k$ and sample-agnostic information through the bias term $\mathbf{b}^k$. This indicates that the learned transition probabilities are conditioned on both the input sample and categorical knowledge. Therefore, the architectural design of the Transition Network complies with (3).

Remark 1: From the given hierarchical labels, we can obtain a deterministic transition matrix $\mathbf{S}^k$, where only one element is 1 in each column. This indicates that $\mathbf{S}_{i,j}^k = 1$ if category j at hierarchy $k-1$ is a subclass of category i at hierarchy k , and $\mathbf{S}_{i,j}^k = 0$ otherwise. Instead, in (13), we aim to model $p(y^k | y^{k-1}, \mathbf{x})$, which can capture data-driven correlation from image context and hierarchy prior. In fact, if we replace $g_{\theta^k}^k(\hat{\mathbf{x}})$ with $\mathbf{S}^k$ in (7) (referred to as LHT_naive), it will result in a significant performance degradation, as the hierarchical relationships derived from a label tree are not always visually grounded. For example, the Aircraft dataset [53], whose hierarchical relationships are based on a *maker-family-model* organization, is defined more by semantics than by visual similarity. As shown in Fig. 3, LHT_naive achieves notably lower performance than our LHT model. To further demonstrate the importance of adaptively learning visual correlations, we constructed a new dataset, Perturbed-Aircraft, which contains a label hierarchy that is less visually consistent than the original dataset Aircraft.² Under this challenging scenario, our LHT model surpasses LHT_naive by 5.1%, further demonstrating the effectiveness of learning data-driven transition matrices to capture visual category correlations.

A lightweight design for $g_{\theta^k}^k$: The architecture of $g_{\theta^k}^k$ designed in (13) requires a large number of parameters. To prevent an explosion of parameters, we propose a simple design for $g_{\theta^k}^k$ as shown in Fig. 4.

²To construct this dataset based on Aircraft, we retain the original three-level hierarchy but randomly perturb 40% of the category indices at the second hierarchy, which introduces significant disorder and misalignment in the category relationships compared to Aircraft.

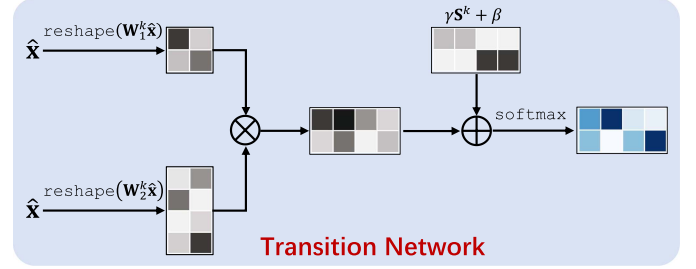


Fig. 4. Architecture of the proposed Transition Network. For the k -th hierarchy, the network includes three small modules: two smaller FC layers with each followed by a reshape operator, i.e., $\text{reshape}(\mathbf{W}_i^k \hat{\mathbf{x}})$, $i = 1, 2$, and one bias offset term $\gamma \mathbf{S}^k + \beta$.

Firstly, we replace the original FC layer with two smaller ones, which means we use a low-rank approximation to estimate $\mathbf{W}^k \hat{\mathbf{x}}$ in (13). This can be formulated as:

$$\text{reshape}(\mathbf{W}^k \hat{\mathbf{x}}) \approx \underbrace{\text{reshape}(\mathbf{W}_1^k \hat{\mathbf{x}})}_{\mathbb{R}^{C_k \times s}} \otimes \underbrace{\text{reshape}(\mathbf{W}_2^k \hat{\mathbf{x}})}_{\mathbb{R}^{C_{k-1} \times s}}, \quad (14)$$

where $\mathbf{W}_1^k \in \mathbb{R}^{(C_k \times s) \times d}$ and $\mathbf{W}_2^k \in \mathbb{R}^{(C_{k-1} \times s) \times d}$ are parameters of the two FC layers, with s denoting a hyper-parameter in order to control the rank of the above approximation. The operation $\mathbf{A} \otimes \mathbf{B}$ denotes the matrix multiplication between $\mathbf{A}$ and $\mathbf{B}^\top$ (i.e., the transpose of $\mathbf{B}$). Eq. (14) indicates that $\mathbf{W}^k \hat{\mathbf{x}}$ is approximated with a rank s matrix, which reduces its parameter complexity from $\mathcal{O}(d(C_k \times C_{k-1}))$ to $\mathcal{O}(ds(C_k + C_{k-1}))$. We initialize s as 2, which ensures efficiency and good performance throughout our experiments. The ablation study on s can be found in Section IV-A-3.

Next, we aim to design a lightweight bias term for replacing $\text{reshape}(\mathbf{b}^k)$ in (13) as it involves a relatively large parameter complexity $\mathcal{O}(C_k \times C_{k-1})$. As previously discussed, this bias term plays a crucial role in capturing sample-agnostic information and encoding categorical knowledge to satisfy (3). Interestingly, the deterministic transition matrix $\mathbf{S}^k$, derived from the given hierarchical labels, can serve as a valuable prior for encoding categorical granularity information. Therefore, we replace $\text{reshape}(\mathbf{b}^k)$ with the following simplified formulation:

$$\text{reshape}(\mathbf{b}^k) \approx \gamma \mathbf{S}^k + \beta, \quad (15)$$

where γ is a scaling parameter to ensure consistency in magnitude between $\gamma \mathbf{S}^k$ and $\mathbf{W}^k \hat{\mathbf{x}}$, and β is a shifting parameter. From (15), we can see that this new bias term introduces negligible parameters compared to the original version.

By substituting (14) and (15) into (13), we obtain a lightweight formulation of the Transition Network: $g_{\theta^k}^k(\mathbf{x}) = \text{softmax}(\text{reshape}(\mathbf{W}^k \hat{\mathbf{x}}) + \text{reshape}(\mathbf{b}^k))$, with the parameters $\theta^k = \{\mathbf{W}_1^k, \mathbf{W}_2^k, \gamma, \beta\}$. We emphasize that our basic principle in designing the transition network is to integrate both sample-specific and sample-agnostic information, as investigated in (3).

Algorithm 1: Mini-Batch Training Algorithm of LHT.

Input: Training data $\mathcal{D}$, hierarchy level K , batch size N , training iteration T , learning rate α .

Output: Parameters of the Classification Network and Transition Networks: $\omega, \theta = \{\theta^k\}_{k=2}^K$.

- 1: Initialize the parameters $\omega^{(1)}$ and $\theta^{(1)}$.
- 2: **for** $t = 1$ **to** T **do**
- 3: Sample a mini-batch of training samples from $\mathcal{D}$:
 $\hat{\mathcal{D}} = \{\mathbf{x}_n, (y_n^1, \dots, y_n^K)\}_{n=1}^N$.
- 4: **for** $k = 1$ **to** K **do**
- 5: **if** $k=1$ **then**
- 6: Compute $\mathbf{p}_n^1$ by (5).
- 7: **else**
- 8: Compute $\mathbf{p}_n^k$ by (6).
- 9: **end if**
- 10: **end for**
- 11: Compute the classification loss $\mathcal{L}_{\text{CE}}$ by (7).
- 12: Compute the confusion loss $\mathcal{L}_{\text{Conf}}$ by (16).
- 13: Compute the overall training loss $\mathcal{L}$ by (17).
- 14: Update
 $\{\omega^{(t+1)}, \theta^{(t+1)}\} = \{\omega^{(t)}, \theta^{(t)}\} - \alpha \nabla_{\omega, \theta} \mathcal{L}$.
- 15: **end for**

Return: Optimized parameters $\omega^{(T)}$ and $\theta^{(T)}$

D. Confusion Loss

As aforementioned in Section III-B, the parameters involved in the Classification Network and the Transition Networks, i.e., w and $\theta = \{\theta^k\}_{k=2}^K$, can be jointly optimized end-to-end using the loss (7). However, the network training by minimizing this objective function remains the following challenges: (i) The manually annotated labels within each hierarchy are one-hot, and such a cross-entropy loss encourages the network to move all categories toward orthogonality at each hierarchy, counteracting the inter-class similarity introduced by class hierarchies; (ii) From the perspective of numerical optimization, there are infinite combinations of $\mathbf{T}^k$ and $\mathbf{p}^{k-1}$, which easily make the learned transition matrices to collapse to a trivial solution.

To address the aforementioned challenges, we further introduce an additional regularization loss on transition matrices $\{\mathbf{T}^k\}_{k=2}^K$, which we call the confusion loss $\mathcal{L}_{\text{Conf}}$. We herein revisit the label hierarchy transition matrices involved in our LHT model. The column vector (e.g., the j -th one) of transition matrix $\mathbf{T}^k$ represents the label distribution of the k -th hierarchy when the sample is labeled as class j at the $(k-1)$ -th hierarchy. Each element of this vector denotes the likelihood of similarity, or confusion, between two classes across two adjacent hierarchies. The proposed confusion loss aims to encourage such class confusion to counter the over-confident prediction induced by the cross-entropy loss. Mathematically, for a batch of N training data $\{\mathbf{x}_n, (y_n^1, y_n^2, \dots, y_n^K)\}_{n=1}^N$, the confusion loss is defined as the negative entropy of the column space spanned by the transition matrices, i.e.,

$$\mathcal{L}_{\text{Conf}} = \sum_{n=1}^N \sum_{k=2}^K \sum_{j=1}^{C_{k-1}} \frac{1}{C_{k-1}} \left\langle \mathbf{T}_n^k[:, j], \log \mathbf{T}_n^k[:, j] \right\rangle, \quad (16)$$

where $\mathbf{T}_n^k$ is the predicted transition matrix for $\mathbf{x}_n$ and $\mathbf{T}_n^k[:, j]$ denotes its j -th column, $\log(\cdot)$ is element-wise logarithm operation for vectors, and $\langle \cdot, \cdot \rangle$ is the inner product operation of vectors. Through minimization of (16), the conditional distribution represented by each column of $\mathbf{T}_n^k$ is encouraged to have a higher entropy (i.e., this distribution inclines to approach uniform distribution). In other words, this introduces *confusion* in fine-to-coarse label transitions during training, thereby encouraging the Transition Networks to encode sample-specific information into transition matrices and alleviating the over-confident prediction induced by the hierarchical cross-entropy loss. Furthermore, since $\mathbf{T}^{k \leftarrow 1}$ can be factorized as $\mathbf{T}^{k \leftarrow 1} = \mathbf{T}^k \dots \mathbf{T}^2$ (Proposition 1), applying confusion loss to each transition matrix may implicitly facilitate the modeling of inter-class similarities between hierarchy 1 and k .

Combined (7) and (16), the overall training loss can be formulated as

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{Conf}}, \quad (17)$$

where λ is hyper-parameter for trade-off between the cross-entropy term $\mathcal{L}_{\text{CE}}$ and the confusion regularization term $\mathcal{L}_{\text{Conf}}$. All the parameters involved in our LHT can be trained in an end-to-end manner. The training algorithm is summarized in Algorithm 1.

IV. EXPERIMENTS ON NATURAL IMAGES

In this section, we first demonstrate the effectiveness of the proposed LHT in **supervised/semi-supervised hierarchical classification** tasks following [29]. Specifically, the labels of training samples are given at all levels of the class hierarchy under the supervised setting. However, under the semi-supervised setting, the labels are observed at different levels of the class hierarchy, indicating that part of samples are only observed at internal levels of the whole hierarchies.³ Next, we demonstrate the applicability of the proposed LHT on **large-scale low-shot learning** where transferable visual features are learned from a class hierarchy to encode the semantic relations between source and target classes. We also provide thorough ablation studies to gain insight into our method by showing the efficacy of each key component.

A. Supervised/Semi-Supervised Hierarchical Learning

1) Experimental Setups. Datasets: In our experiments, we evaluate our method on three public benchmark datasets: CUB-200-2011 [18], Aircraft [53] and Stanford Cars [54], where the label hierarchies are publicly available or constructed by Chang et al. [27] according to their lexical relationships in Wikipedia. The details of these datasets are as follows:

- **CUB-200-2011**⁴ is a dataset of bird images mainly introduced for the problem of fine-grained visual categorization methods. It contains 11,877 images belonging to

³In this paper, we stem the term of “semi-supervised learning” from [29], in the sense of “partially supervised with incomplete hierarchical labels for a part of training samples”, in contrast to the classical term that assumes most training samples are given without any annotation.

⁴<http://www.vision.caltech.edu/visipedia/CUB-200-2011.html>

200 bird species. For hierarchical classification evaluation, the labels are re-organized into three-level hierarchy with 13 orders, 38 families and 200 species by tracing their biological taxonomy [27]. For example, the original label “*Brewer Blackbird*” is spanned into (“*Passeriformes*”, “*Icteridae*”, “*Brewer Blackbird*”) in the order of (Order, Family, Species).

- **Aircraft**⁵ consists of 10,000 images across 100 airplane classes, with a data ratio of approximately 2:1 between the training and test sets. The dataset includes a three-level label hierarchy with 30 makers (e.g., “*Boeing*”), 70 families (e.g., “*Boeing 767*”) and 100 models (e.g., “*767-200*”). As a result, this dataset can be readily used in our experiments without any modification.
- **Stanford Cars**⁶ contains 8,144 training and 8,041 test images across 196 car classes. It is re-organized into two-level label hierarchy, i.e., 9 car types and 196 car classes as introduced in [27]. For example, the class label “*BMW X6 SUV 2012*” are spanned into (“*SUV*”, “*BMW X6 SUV 2012*”). We use the standard public train/test splits, and not any bounding box/part annotations are involved in all our experiments.

For the supervised hierarchical classification setting, we train the model using the complete set of labels at all hierarchical levels for each training sample. For the semi-supervised hierarchical classification setting, in order to simulate a lack of domain knowledge, we randomly select 30%, 50%, 70%, and 90% samples (denoted as $\mathcal{R} = 90\%$ for example) to mask out their labels at the finest-level hierarchy, such that their intermediate parent classes are relabeled as the leaf classes. When employing this approach in semi-supervised scenarios, in our model training process, we simply neglect the loss imposed on the finest-level hierarchy for the samples that have been relabeled to their parent classes.

Evaluation Metrics: We adopt the same evaluation metrics as Chang et al. [27] to evaluate our method. Specifically, *ACC* (i.e., accuracy, the percentage of images whose labels are correctly classified) is reported at each class hierarchy, and then *mACC* (the mean of *ACC* across all class hierarchies) is calculated to measure the classification performance at a hierarchical level. Note that, unless otherwise specified, the *ACC* results are reported as the mean over three random seeds of each experiment.

Implementation Details: All our experiments are implemented with the Pytorch platform [56] running on NVIDIA Geforce RTX 3090Ti GPUs. Following [27], we employ ResNet-50 pre-trained on ImageNet [57] as our network backbone, followed by a fully-connected layer with 600 hidden units. For fair comparison, the features of fully-connected layer are uniformly split into K parts ($K = 2$ for Standford Cars and $K = 3$ for other datasets), with each used for predicting the labels at each hierarchy. During training, the models are trained with Momentum SGD with a momentum of 0.9, a weight decay of 5×10^{-4} , a batch-size of 8, and an initial learning rate of 0.0002 for backbone layers and 0.002 for other layers. The

training epoch is set to be 200 and the learning rate is adjusted by the cosine annealing schedule. We initially set $\lambda = 0.1$ for Cub-200-2011 and $\lambda = 2.0$ for the other two datasets, except for $\lambda = 0.01$ for all the three datasets under a relabeled ratio of $\mathcal{R} = 90\%$. The same augmentation strategies as described in [29] are adopted, i.e., each image is resized to 448×448 , and then the resulting image is randomly cropped (random cropping for training and center cropping for testing) and randomly horizontally flipped.

2) **Main Results:** We compare the proposed method with state-of-the-art methods on this task, including: (i) **HD-CNN** [24], which aims to learn a hierarchical deep convolutional neural network where a single network backbone was shared by multiple fully-connected layers with each one responsible for the label prediction within one hierarchy; (ii) **HMCN** [55], which designs a specific network architecture with multiple local outputs for label prediction per hierarchical level and one global output layer for capturing semantic correlation in the hierarchy as a whole; (iii) **C-HMCNN** [46], which exploits the hierarchy information by imposing the parent-child constraint on the output of the coarser-level hierarchies, so as to ensure that no hierarchy violation happens (i.e., when the network predicts a sample belonging to a class, this sample also belongs to its parent classes); (iv) **Soft Labels** [17], an embedding-based method that softens the finest-level labels according to the metric derived by the label tree during training, and during testing, we backtrack through the label tree to obtain other coarse-level labels; (v) **FGoN** [27], one of the current state-of-the-arts which equips with hierarchy-specific classification head to disentangle and reinforce the features of different hierarchies, that is, only forces finer-level features to participate in coarser-level label predictions; (vi) **HRN** [29], which incorporates hierarchical feature interaction into model design and proposes a hierarchical residual network, where hierarchy-specific features from parent level are added to features of children levels through residual connections. For fair comparison, we conduct all experiments under a unified codebase that includes: the public source codes of C-HMCNN [46],⁷ Soft Labels [17],⁸ FGoN [27]⁹ and HRN [29],¹⁰ and the reproduced codes of HD-CNN [24] and HMCN [55] since no source code is available for the original papers. When adapting these methods to semi-supervised settings, we neglect the loss over the leaf class if a sample has been relabeled to its parent class.

In Tables I, II and III, we report ACC of each hierarchical level and mACC results on test sets of CUB-200-2011, Aircraft, and Stanford Cars, respectively, obtained by all competing methods. For **supervised hierarchical classification** setups (i.e., $\mathcal{R} = 0\%$), we can observe that our LHT model consistently achieves superior or comparable performance in terms of ACC at each hierarchy, and it consistently outperforms all compared methods over mACC on all three datasets. For **semi-supervised hierarchical classification**, we select four setups

⁷<https://github.com/EGiunchiglia/C-HMCNN>

⁸github.com/fiveai/making-better-mistakes

⁹<https://github.com/PRIS-CV/Fine-Grained-or-Not>

¹⁰<https://github.com/MonsterZhZh/HRN>

⁵<https://www.robots.ox.ac.uk/vgg/data/fgvc-aircraft>

⁶<https://www.kaggle.com/datasets/jessicali9530/stanford-cars-dataset>

TABLE I

COMPARISON RESULTS OF ALL COMPETING METHODS OVER ACC AND mACC ON CUB-200-2011 DATASET. R DENOTES THAT A FRACTION OF $R\%$ OF THE TRAINING SAMPLES LACK LABELS AT THE FINEST-LEVEL (SPECIES) HIERARCHY. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**. * INDICATES THE RESULTS REPORTED IN THE ORIGINAL PAPER.

R	Hierarchy	HD-CNN [24]		HMCN [55]		C-HMCNN [46]		Soft Labels [17]		FGoN [27]		HRN [29]		HRN [29]*		Ours	
		ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC
0%	Order	98.5		98.2		98.6		98.8		98.8		98.8		98.7		99.0	
	Family	95.5	93.3	94.7	93.0	95.7	93.6	95.9	93.7	95.5	93.6	95.5	93.7	95.5	93.6	96.2	94.1
	Species	85.9		86.0		86.5		86.5		86.6		86.7		86.6		87.0	
30%	Order	98.5		98.0		98.5		98.9		98.7		98.5		98.3		98.9	
	Family	95.3	92.5	94.0	91.9	95.2	92.8	95.7	93.1	95.2	93.0	95.0	92.4	94.8	92.3	96.2	93.7
	Species	83.8		83.6		84.6		84.8		85.1		83.6		83.9		86.0	
50%	Order	98.4		97.9		98.5		98.6		98.7		98.0		97.9		98.9	
	Family	94.7	91.7	93.5	90.2	94.8	91.9	95.0	92.1	95.3	92.3	94.7	91.3	94.3	90.9	96.1	93.3
	Species	81.9		79.2		82.3		82.7		82.8		81.1		80.5		84.8	
70%	Order	98.4		97.7		96.9		97.6		98.7		97.2		98.4		98.9	
	Family	94.7	90.1	92.7	87.1	91.3	87.8	92.6	89.3	94.9	90.7	93.3	88.3	93.9	88.8	96.0	92.4
	Species	77.2		70.9		75.2		77.8		78.6		74.4		74.0		82.4	
90%	Order	98.5		97.8		91.2		94.1		98.7		97.7		98.0		98.8	
	Family	94.6	83.8	90.6	79.3	77.1	72.6	83.4	79.1	94.3	85.6	93.6	83.4	93.3	81.4	95.3	87.6
	Species	58.3		49.4		49.6		59.7		63.8		58.8		53.0		68.8	

TABLE II

COMPARISON RESULTS OF ALL COMPETING METHODS OVER ACC AND mACC ON AIRCRAFT DATASET. R DENOTES THAT A FRACTION OF $R\%$ OF THE TRAINING SAMPLES LACK LABELS AT THE FINEST-LEVEL (MODEL) HIERARCHY. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**. * INDICATES THE RESULTS REPORTED IN THE ORIGINAL PAPER.

R	Hierarchy	HD-CNN [24]		HMCN [55]		C-HMCNN [46]		Soft Labels [17]		FGoN [27]		HRN [29]		HRN [29]*		Ours	
		ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC
0%	Maker	96.9		97.2		97.1		97.1		97.1		97.3		97.5		97.4	
	Family	94.9	94.3	95.6	95.1	95.4	94.6	95.0	94.4	95.5	94.7	95.7	95.2	95.8	95.3	95.9	95.3
	Model	91.1		92.6		91.4		91.1		91.4		92.7		92.6		92.6	
30%	Maker	96.9		97.2		96.3		96.8		96.9		97.3		97.3		97.2	
	Family	94.6	93.7	95.6	94.8	94.0	93.1	94.1	93.3	95.4	94.1	95.3	94.5	95.5	94.8	95.8	95.0
	Model	89.7		91.6		89.0		89.1		90.0		91.0		91.6		91.9	
50%	Maker	96.8		96.9		95.9		96.0		96.9		96.8		97.3		97.3	
	Family	94.8	93.4	95.1	93.8	93.2	92.1	93.0	91.9	95.3	93.5	95.1	94.0	95.7	94.2	95.8	94.8
	Model	88.7		89.3		87.1		86.6		88.2		90.1		89.7		91.4	
70%	Maker	96.7		96.9		93.4		94.1		96.9		96.2		96.8		97.3	
	Family	94.5	92.3	94.4	92.1	89.7	88.0	89.8	88.0	95.2	92.0	93.8	91.9	94.2	91.8	95.3	93.6
	Model	85.8		84.9		80.8		80.0		83.8		85.7		84.5		88.1	
90%	Maker	96.6		96.9		73.7		82.4		96.4		96.3		95.4		97.1	
	Family	93.8	89.0	94.7	89.1	63.0	61.6	72.4	71.0	94.6	87.6	92.5	87.3	91.7	86.1	95.0	90.5
	Model	76.7		75.8		48.1		58.2		71.9		73.2		71.1		79.3	

TABLE III

COMPARISON RESULTS OF ALL COMPETING METHODS OVER ACC AND mACC ON STANFORD CARS DATASET. R DENOTES THAT A FRACTION OF $R\%$ OF THE TRAINING SAMPLES LACK LABELS AT THE FINEST-LEVEL (MAKER) HIERARCHY. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**. * INDICATES THE RESULTS REPORTED IN THE ORIGINAL PAPER.

R	Hierarchy	HD-CNN [24]		HMCN [55]		C-HMCNN [46]		Soft Labels [17]		FGoN [27]		HRN [29]		HRN [29]*		Ours	
		ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC	ACC	mACC
0%	Type	96.8		97.0		97.7		97.2		96.7		97.2		97.4		97.0	
	Maker	93.0	94.9	93.4	95.2	93.1	95.4	93.5	95.4	93.4	95.1	93.6	95.4	94.0	95.7	94.1	95.6
30%	Type	96.5		96.6		96.9		97.1		96.5		96.5		96.1		97.0	
	Maker	91.8	94.2	91.4	94.0	90.9	93.9	92.0	94.6	91.8	94.2	91.4	94.0	90.6	93.4	93.4	95.2
50%	Type	96.3		96.6		95.5		96.1		96.2		96.0		95.9		96.9	
	Maker	89.7	93.0	87.6	92.1	88.4	92.0	89.8	93.0	90.0	93.1	88.3	92.2	88.7	92.3	92.7	94.8
70%	Type	96.1		96.2		93.4		93.9		95.8		96.0		96.1		96.9	
	Maker	84.8	90.5	77.3	86.8	80.0	86.7	83.7	88.8	84.0	89.9	79.7	87.9	83.7	89.9	89.9	93.4
90%	Type	95.3		95.2		82.4		80.0		95.0		94.1		94.3		96.0	
	Maker	57.8	76.6	44.4	69.8	48.5	65.5	55.1	67.6	59.7	77.4	47.9	71.0	49.3	71.8	71.4	83.7

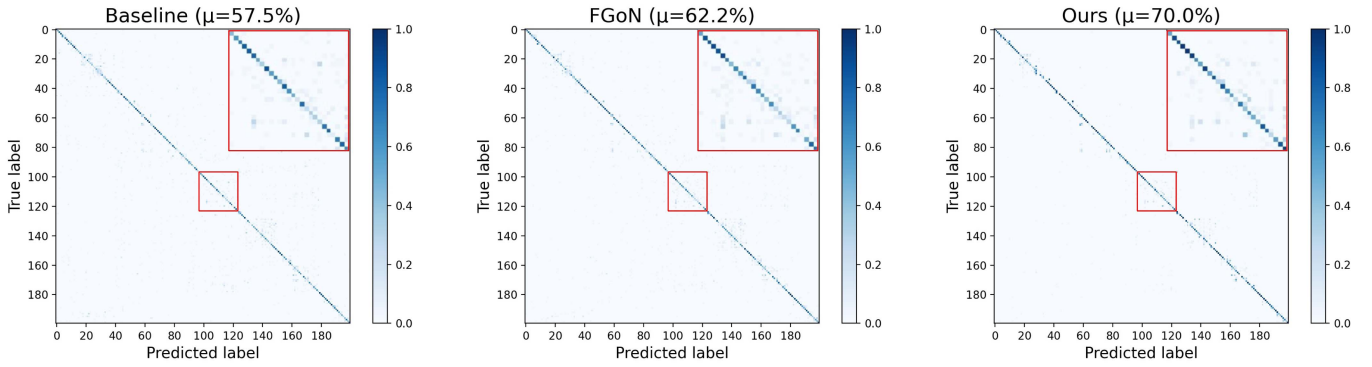


Fig. 5. Confusion matrices of Baseline, FGoN [27] and our proposed LHT method (Ours) at the finest-grained hierarchy on Cub-200-2011 with the relabeled ratio $\mathcal{R} = 90$. Here μ above each figure represents mean-class-recall for all 200 classes.

(i.e., $\mathcal{R} = 30\%, 50\%, 70\%, 90\%$) to imitate the lack of domain knowledge. Note that the dataset with $\mathcal{R} = 30\%$ means that we randomly select 30% samples from the training set and mask out their labels at the finest-level hierarchy during training. It can be observed that: 1) Our LHT achieves the best or second best ACC and mACC performance across all the hierarchies on all the three datasets. Particularly, in cases where limited fine-grained label information ($\mathcal{R} = 90\%$) is available, our LHT surpasses all compared methods at all hierarchies by an evident margin, resulting in absolute mACC improvements of 5.0%, 2.6%, and 13.6% compared to the second best results on all three datasets, respectively. 2) The performance of all compared methods degrade with an increasing number of relabeled samples ($\mathcal{R} = 0\% \rightarrow 90\%$), but our LHT model consistently outperforms current state-of-the-art methods. This indicates that LHT can better utilize hierarchical label information to learn a more discriminative classifier. 3) To elucidate the origins of performance improvements, we visually represent the confusion matrices obtained from the test set of Cub-200-2011 dataset, employing a relabeled ratio of $\mathcal{R} = 90\%$. It should be noted that the diagonal vector of a confusion matrix signifies per-class recall. As shown in Fig. 5, our LHT model exhibits a notably superior improvement compared to the baseline ResNet-50 and the current state-of-the-art method FGoN [27]. Specifically, LHT achieves a mean-class-recall (defined as the arithmetic mean of recalls across all 200 classes) of 70.0%, surpassing FGoN by 7.8%. These results substantiate the superiority of our LHT in effectively capturing hierarchical knowledge during the training process.

3) *Ablation Study*: In order to gain a deeper understanding of the effectiveness underlying the proposed LHT model, We perform extensive ablation study to evaluate and understand the contribution of each critical component in the proposed LHT model. Unless mentioned otherwise, the experiments in this subsection are all performed on Cub-200-2011 dataset with relabeled ratio $\mathcal{R} = 90\%$.

First, we investigate the impact of label hierarchy to demonstrate that hierarchical label information is beneficial for performance enhancement, but it requires effective learning mechanisms to exploit its underlying semantics. To this end, we append a branch for each hierarchy level on ResNet-50 and use the

TABLE IV
ABLATION STUDY ON EACH KEY COMPONENT OF OUR LHT FRAMEWORK. THE PERFORMANCE (ACC) IS EVALUATED AT THE FINEST-LEVEL (SPECIES) HIERARCHY BY GRADUALLY ADDING HIERARCHICAL CROSS-ENTROPY LOSS (HCL), TRANSITION NETWORKS, AND CONFUSION LOSS INTO THE BACKBONE RESNET-50.

Model	ACC
ResNet-50	55.9
ResNet-50 + HCL	58.3
ResNet-50 + HCL + Transition Networks	66.7
LHT (Ours)	68.8
LHT_simplified (w/ Trainable Transition Matrices)	58.1

hierarchical cross-entropy loss (denoted as ResNet-50 + HCL) to fit the labels at each hierarchy, and the results are listed in Table IV. It can be observed that the performance only exhibits a marginal improvement of about 2.4%, indicating a limited boost in comparison to the advancements achieved by our proposed LHT model.

Next, we validate the superiority of the proposed transition network, which aims to effectively encodes visual correlations embedded in class hierarchies, as delineated in Section III-C. To substantiate this, on one hand, we quantify the performance of the model by integrating our proposed Transition Networks into the baseline model (ResNet-50 + HCL). As shown in Table IV, the model achieves an accuracy of 66.7%, which yields an absolute gain of 8.4% over the baseline model. This substantiates the efficacy of our proposed transition network. We further investigate the hyper-parameter s introduced by the Transition Networks in Fig. 6(a). We can see that the performance remains relatively stable as the value of s varies. As such, we fix it as 2 across all our experiments for a better trade-off between computation efficiency and model performance. On the other hand, if we directly treat the transition matrices as trainable parameters (LHT_simplified) instead of learning them through the Transition Networks, model performance degrades from 68.8% to 58.1%, further validating the effectiveness of the Transition Networks.

Lastly, we conduct ablation analysis on the proposed confusion loss. As shown in Table IV, when the confusion loss

TABLE V
PERFORMANCE COMPARISON ACROSS DIFFERENT BACKBONES ON CUB-200-2011. WE REPORT RESULTS ON THREE REPRESENTATIVE CNN AND ViT ARCHITECTURES: RESNET-50, CONVNEXT-BASE, AND SWIN-BASE. WE OMIT THE “BASE” SUFFIX IN THE TABLE.

Method	Backbone	$\mathcal{R} = 0$			$\mathcal{R} = 30$			$\mathcal{R} = 50$			$\mathcal{R} = 70$			$\mathcal{R} = 90$		
		Order	Family	Species	Order	Family	Species	Order	Family	Species	Order	Family	Species	Order	Family	Species
Baseline	ResNet-50	98.5	95.5	85.9	98.5	95.3	83.8	98.4	94.7	81.9	98.4	94.7	77.2	98.5	94.6	58.3
Ours	ResNet-50	99.0	96.2	87.0	98.9	96.2	86.0	98.9	96.1	84.8	98.9	96.0	82.4	98.8	95.3	68.8
Baseline	ConvNeXt	99.2	96.5	89.2	99.2	96.4	88.3	99.2	96.3	87.0	99.1	96.0	83.9	99.0	96.0	73.3
Ours	ConvNeXt	99.4	97.4	89.3	99.4	97.3	88.8	99.4	97.4	87.8	99.3	97.1	85.7	99.3	96.5	80.1
Baseline	Swin	99.2	96.7	89.2	99.1	96.6	88.3	99.1	96.6	86.5	99.1	96.4	83.4	99.0	95.9	72.4
Ours	Swin	99.5	97.4	89.5	99.5	97.2	88.7	99.5	97.2	87.5	99.5	97.1	85.5	99.3	96.6	78.9

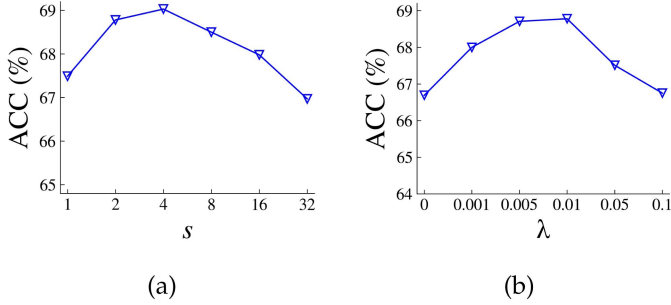


Fig. 6. Ablation study on two hyper-parameters: (a) s in the transition network (14) and (b) λ in the training loss (17).

is integrated into the model (ResNet-50 + HCL + Transition Networks), the resulting LHT model achieves a further 2.1% performance improvement. This verifies the effectiveness of the proposed confusion loss on encouraging the transition network to capture the semantic correlation within class hierarchies. In fact, the hyper-parameter λ in (17) plays an important role in trading off between the hierarchical cross-entropy loss and the confusion loss during training, and we thus conduct a detailed ablation analysis on λ in Fig. 6(b). It can be observed that the initial performance gradually improves as λ increase from 0 to 0.01, which suggests that a smaller λ facilitates the transfer network to output meaningful transfer matrices; however, a much larger λ excessively forces the output transfer matrices to be more smooth, which may amplify biases introduced by inaccurate predictions, especially in semi-supervised scenarios.

4) *Discussion and Analysis:* We provide a comprehensive discussion from multiple perspectives to enrich the understanding of our approach by addressing the following questions.

How does LHT generalize across various architectures? In addition to ResNet-50, we further evaluate our method on ConvNeXt-Base and Swin-Base, which are representative modern CNN and ViT architectures. Compared to ResNet-50, these models have substantially larger parameter capacities, extending the evaluation to a much broader capacity range. We keep all other experimental settings consistent with those used for ResNet-50. As shown in Table V, our proposed LHT consistently outperforms the baseline HD-CNN [24] that employs independent classification heads for each hierarchy under varying missing-label ratios. These results demonstrate that our approach generalizes well to both advanced CNNs and ViT-based architectures, confirming its robustness across diverse architecture designs.

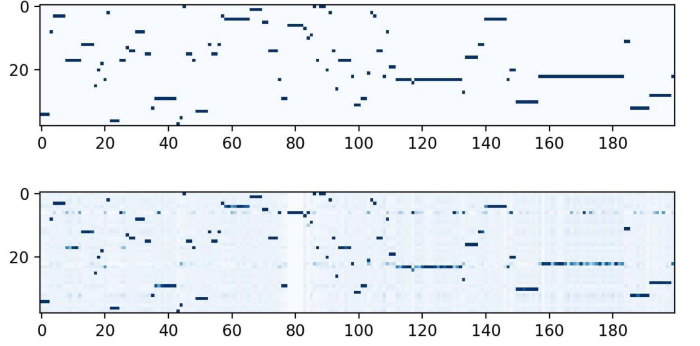


Fig. 7. Visualization of transition matrices (*Species* $\rightarrow$ *Family*) on CUB-200-2011: the top shows the naive transition matrix S^1 inferred from hierarchical labels, and the bottom shows the learned transition matrix T^1 .

How does LHT perform under sparser annotation settings? As aforementioned, our LHT presents an effective learning mechanism to exploit hierarchical label information, especially in cases where a significant portion of labels are missing at the finest-level hierarchy. Herein, we further validate the efficacy of our method in a more challenging semi-supervised scenario, where we randomly select a fraction of samples and mask out their labels at the two finest-level hierarchies (i.e., Species and Family) on Cub-200-2011 dataset. As shown in Table VI, Our proposed LHT model consistently improves the baseline HD-CNN [24] across different amounts of relabeled data. For example, in an extremely sparse annotation scenario ($R^2 = 90\%$), LHT significantly improves the baseline model by around 9.4% ACC at the finest-level hierarchy and 4.3% mACC across all hierarchies. This demonstrates that our LHT achieves higher performance with less label information, thus confirming its superiority in effectively utilizing class hierarchies.

Does the learned transition matrix encode semantic correlations? To answer this question, we visualize the transition matrices output by the Transition Network on Cub-200-2011 in Fig. 7. Specifically, we show the learned transition matrix associated with the two finest-level hierarchies, denoted as $T^1(\text{species} \rightarrow \text{family})$, and also visualize the corresponding naive transition matrices $S^1(\text{species} \rightarrow \text{family})$ directly inferred from hierarchical labels (as mentioned in Remark 1) for better reference. We can see that the naive transition matrix can only encode the deterministic state transition from the finer- to the coarser-level hierarchies since only one element is 1 in each column, whereas the ones learned from our proposed LHT model are indeed conditional label distributions considering the uncertainty from

TABLE VI
ANALYSIS FOR MORE SPARSE ANNOTATION SETTINGS ON CUB-200-2011 DATASET. R^2 DENOTES THAT A FRACTION OF $R\%$ OF THE TRAINING SAMPLES LACK LABELS AT THE TWO FINEST-LEVEL (SPECIES AND FAMILY) HIERARCHIES.

Method	$\mathcal{R} = 30$			$\mathcal{R} = 50$			$\mathcal{R} = 70$			$\mathcal{R} = 90$		
	Order	Family	Species	Order	Family	Species	Order	Family	Species	Order	Family	Species
Baseline	98.5	94.5	83.8	98.4	93.4	81.4	98.4	91.1	75.2	97.9	82.4	51.0
Ours	98.9	95.9	85.3	98.9	95.6	84.3	98.8	94.8	81.1	98.3	85.6	60.4

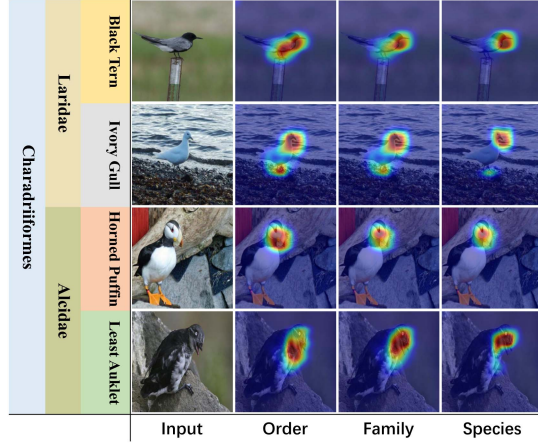


Fig. 8. Visualization of the attention maps returned by Grad-Cam [58] of the three hierarchical branches of the proposed LHT model, with each one corresponding to a specific hierarchy. The leftmost column is the original image, and the three columns on the right correspond to the attention maps of Order, Family and Species hierarchy, respectively.

image context and hierarchy prior. This is mainly because, on the one hand, the learned matrices largely reflect the transition states in the naive transfer matrix, while on the other hand, LHT produces distinct transition matrices for different input samples. We also visualize the evolution of the weight γ during training in Appendix E, available online, to clarify the impact of the learned transition matrices.

Does the learned attention map encode semantic correlations? In Fig. 8, we further employ Grad-Cam [58] to visualize the attention regions of each hierarchical branch by propagating their respective gradients back to feature maps for four typical examples. These objects belong to the same order (i.e., “Charadriiformes”), two families (i.e., “Laridae” and “Alcidae”), and four species (i.e., “Black Tern”, “Ivory Gull”, “Least Auklet” and “Horned Puffin”). It can be observed that 1) our proposed LHT model exhibits a preference for activating more global attention regions in the coarser-level hierarchy compared to the finer-level hierarchy, and the visualization results demonstrate notable difference between the attention patterns of the finer-level categories and that of the coarser-level categories; 2) our LHT model effectively directs attention towards discriminative regions when classifying examples from different species within the same family. For example, when distinguishing between the species “Least Auklet” and “Horned Puffin”, both belonging to the family “Alcidae”, the model focuses intensely on the head, particularly the eye and beak regions, which provide essential discriminative information. This further validates the

capability of our LHT model in capturing semantic correlations across different hierarchies to some extent.

B. Low-Shot Learning With Class Hierarchy

Large-scale low-shot learning (LSL) involves a large number of source classes with abundant labeled data and target classes with only a few labeled examples. The goal is to recognize unlabeled target samples, which is challenging due to their limited supervision and the difficulty of directly learning discriminative classifiers.

1) *Problem Formulation:* Formally, let S_{source} denote the set of source classes and S_{target} denote the set of target classes. The two class sets satisfy the following properties: 1) S_{source} and S_{target} are disjoint, i.e., $S_{source} \cap S_{target} = \emptyset$; 2) the semantic correlation of S_{source} and S_{target} are explicitly encoded into a tree-structured class hierarchy, where the leaf nodes cover all source and target classes and the parent nodes group semantically related leaf classes. We are given a large-scale source set $\mathcal{D}_{source}$ from source classes S_{source} , a few-shot support set $\mathcal{D}_{target}$ from target classes S_{target} , and a test set $\mathcal{D}_{test}$ from target classes S_{target} . In this setting, $\mathcal{D}_{source}$ contains sufficient labeled samples for each source class, while $\mathcal{D}_{target}$ provides only a few (e.g., 5) labeled samples per target class. The objective is to predict labels for $\mathcal{D}_{test}$.

Following [59], we address the challenge of LSL by introducing hierarchical classification. The key idea is that, although source and target classes are disjoint, they often share higher-level categories in a tree-structured hierarchy. For example, certain source and target classes may share the broader category of “vehicles”, with the former covering various vehicle types and the latter including specific categories such as cars. This shared structure provides a natural bridge between the source and target classes. By performing a hierarchical classification on the source data, the network learns transferable features that capture such semantic relationships. These representations serve as informative priors from semantically related source classes and are particularly beneficial for recognizing target classes.

Specifically, the method follows a two-stage process: i) Training stage, in which only source class samples are used to train the hierarchical network. Since the source classes and target classes share certain common super-classes, the network facilitates the learning of transferable features for LSL on the target classes. ii) Test stage, in which visual features of the few-shot support samples and the test samples (both from target classes) are extracted through the learned hierarchical classification network. The test samples are then recognized by a simple nearest neighbor search using the visual features of the support samples as references.

TABLE VII
RESULTS FOR LARGE-SCALE LSL ON ILSVRC2012/2010 DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Methods	K -shot				
	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$
NN	34.2	43.6	48.7	52.3	54.0
SGM [61]	31.6	42.5	49.0	53.5	56.8
PPA [62]	33.0	43.1	48.5	52.5	55.4
LSD [63]	33.2	44.7	50.2	53.4	57.6
KT [59]	39.0	48.9	54.9	58.7	60.5
Ours	41.9	52.2	58.6	61.6	63.7

Specifically, when adapting the proposed LHT method to large-scale LSL, we train the hierarchical classification network with the source data in the training stage and employ the features from global average pooling as input for the nearest-neighbor classifier in the test stage.

2) *Experimental Setups*: A benchmark derived from ILSVRC2012/2010 [57] is used for performance evaluation [59]. This dataset consists of three parts: a support set with 1000 source classes from ILSVRC2012, a few-shot support set with 360 target classes from ILSVRC2010 (not included in ILSVRC2012), and a test set containing target class samples excluding those in the support set. Following [59], we construct a class hierarchy with three levels according to: 1) we first cluster the word vectors¹¹ of source/target class names using k-means to create a coarser-level hierarchy, and the super-class word vectors are obtained by averaging the word vectors of their child classes; 2) we then acquire upper-layer nodes by clustering the word vectors of the lower-layer nodes. These three hierarchies have 1000, 100, and 10 classes, respectively. We refer to [59] and employ Momentum SGD with a momentum of 0.9, a weight decay of 5×10^{-4} , a batch-size of 128, and an initial learning rate of 0.001 for backbone layers and 0.01 for other layers to train the model for 5 epochs. The top-5 accuracy on $\mathcal{D}_{test}$ is adopted as the evaluation metric.

3) *Experimental Results*: We compare our model with five large-scale LSL models: (i) NN, a baseline model employing ResNet50 pre-trained on $\mathcal{D}_{source}$ as the nearest neighbor (NN) classifier; (ii) SGM [61], a LSL model using the squared gradient magnitude (SGM) loss [7]; (iii) PPA [62], the parameter prediction with activation (PPA) model; (iv) LSD [63], a LSL model with large-scale diffusion (LSD); (v) KT [59], the state-of-the-art LSL model via knowledge transfer (KT) with class hierarchy. The results in Table VII show that our proposed LHT method achieves the best result among all the comparison methods, and the performance gain is significant compared to KT. While KT also introduces hierarchical classification by feeding fine-grained predictions into coarser-level classifiers, it does not explicitly model statistical dependencies across hierarchies. Instead, our LHT makes these dependencies explicit by formulating inter-level correlations as conditional probabilities,

¹¹The word vectors are obtained by a skip-gram text model [60] in [59] and available at <https://github.com/tiangelo/fsl-hierarchy>.

TABLE VIII
SUMMARY OF THE ISIC2018 DATASET. THE SKIN LESION CATEGORIES ARE ORGANIZED INTO THREE HIERARCHICAL LEVELS.

Class Hierarchy	Melanocytic		Non-melanocytic				
	Benign	Malignant	Benign		Malignant		
	NV	MEL	BKL	VASC	DF	BCC	AKIEC
Number	6705	1113	1098	143	115	514	327

which facilitates the capture of hierarchical relationships during training. This design yields more transferable features and explains the consistent improvements compared to KT.

V. EXPERIMENTS ON MEDICAL IMAGES

In this section, we empirically shed light on the potential of the proposed LHT applied in the domain of computer-aided diagnosis. Concretely, we reformulate the lesion diagnosis problem into a hierarchical classification task, as we can readily obtain a hierarchical categorization of lesions based on pathological priors [36], [64].

A. Experimental Setups

Dataset: The proposed model is evaluated on the publicly available ISIC2018 dataset [65] from the ISIC2018 challenge on skin lesion diagnosis.¹² It is a challenging dataset due to the class-imbalanced issue. The dataset contains totally 10015 dermoscopic images for seven skin diseases, namely Melanocytic Nevi (NV), Melanoma (MEL), Benign Keratosis (BKL), Basal Cell Carcinoma (BCC), Actinic Keratosis / Bowens disease (AKIEC), Vascular Lesion (VASC), and Dermatofibroma (DF). We organize these seven skin lesions into three hierarchies according to their origin (melanocytic or non-melanocytic), degree of malignancy (benign or malignant), and finally their diagnosis (e.g., melanoma, basal cell carcinoma, vascular lesion, etc.) following [64], [66]. In Table VIII, we list the statistics and class hierarchies of the dataset. Since the ground truth of official validation and testing set was not released, we randomly split the entire training set to 70% for training, 10% for validation and 20% for testing. To verify the proposed method can notably improve the performance of skin lesion diagnosis and concurrently reduce the annotation cost of training samples, we conduct experiments in a semi-supervised learning regime, i.e., randomly selecting 80% of training samples and masking out their labels at the finest-level hierarchy ($\mathcal{R} = 80\%$) or the two finest-level hierarchies ($\mathcal{R}^2 = 80\%$).

Evaluation Metrics: To quantitatively evaluate the proposed method, we adopt five metrics, including Precision, Recall, F1-score, AUC¹³ and Accuracy for evaluation. We report the averaged results from five independent runs with different random seeds.

Implementation details: We employ ResNet-50 pre-trained on ImageNet [57] as our network backbone, and train the models

¹²<https://challenge2018.isic-archive.com>

¹³Each metric is calculated as the average value across all classes.

TABLE IX

PER-CLASS RECALL (%) ON THE TEST SET OF ISIC2018 DATASET. R AND R^2 REPRESENT THE RANDOM SELECTION OF $R\%$ TRAINING SAMPLES FOR LABEL MASKING AT THE FINEST-LEVEL AND TWO FINEST-LEVEL HIERARCHIES, RESPECTIVELY.

Method	Training Data		Lesion Category							Avg.
	Labeled	Partially-labeled	NV	MEL	BKL	BCC	AKIEC	VASC	DF	
ResNet-50	20%	0	94.1	42.4	60.1	68.7	68.7	77.9	58.2	67.2
LHT (Ours)	20%	$\mathcal{R} = 80\%$	97.1	63.3	80.4	87.7	78.8	89.7	80.0	82.4
LHT (Ours)	20%	$R^2 = 80\%$	96.4	42.7	74.1	81.0	77.6	86.2	72.9	75.8

TABLE X

COMPARISON RESULTS OVER PRECISION, RECALL, ACCURACY, F1-SCORE AND AUC ON ISIC2018 DATASET. R AND R^2 REPRESENT THE RANDOM SELECTION OF $R\%$ TRAINING SAMPLES FOR LABEL MASKING AT THE FINEST-LEVEL AND TWO FINEST-LEVEL HIERARCHIES, RESPECTIVELY. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND UNDERLINE, RESPECTIVELY.

Method	ISIC2018 ($\mathcal{R} = 80\%$)					ISIC2018 ($R^2 = 80\%$)				
	Precision	Recall	Accuracy	F1-Score	AUC	Precision	Recall	Accuracy	F1-Score	AUC
HD-CNN [24]	79.8	<u>79.9</u>	<u>89.8</u>	<u>79.4</u>	97.3	73.4	73.0	84.9	72.3	95.1
HMCN [55]	78.5	72.2	86.6	74.3	94.5	66.6	48.4	78.0	53.7	85.9
C-HMCNN [46]	<u>80.2</u>	78.2	89.4	78.8	<u>97.5</u>	<u>73.6</u>	72.4	<u>85.4</u>	72.1	<u>95.8</u>
FGoN [27]	78.8	78.2	88.5	77.8	96.9	<u>72.9</u>	<u>73.7</u>	84.9	<u>72.7</u>	95.1
HRN [29]	77.7	77.6	88.6	77.1	96.4	72.8	71.5	83.9	71.8	94.3
LHT (Ours)	81.3	82.4	90.1	81.3	98.2	76.4	75.8	86.1	75.3	96.7

with Momentum SGD with a momentum of 0.9, a weight decay of 5×10^{-4} , a batch-size of 8, and an initial learning rate of 0.0002 for backbone layers and 0.002 for other layers. The training epoch is set as 200 and the learning rate is decayed by the cosine annealing schedule. The hyper-parameter λ in (17) is set as 0.01. In all our experiments, we resize input images to 448×448 (originally around 600×450 pixels) and simply apply random horizontal flipping and random cropping (random cropping for training and center cropping for testing) for data augmentation.

B. Experimental Results

To understand the effectiveness of the proposed LHT applied to skin lesion diagnosis, we herein answer the following research questions under both $\mathcal{R} = 80\%$ and $R^2 = 80\%$ settings: 1) Does LHT contribute to the diagnosis of skin lesions by using class hierarchy? 2) How does LHT perform compared with prior hierarchical classification methods when used in skin lesion diagnosis?

To answer the first question, we report the per-class recall results for all the seven classes and compare it with a vanilla model (ResNet-50 without using hierarchical label information to train the model) in Table IX. It can be observed that: 1) Our proposed LHT method consistently improves the recall of each lesion category and improve the average recall by $67.2\% \rightarrow 82.4\%$ under $\mathcal{R} = 80\%$ and $67.2\% \rightarrow 75.8\%$ under $R^2 = 80\%$, respectively. This demonstrates that our approach effectively leverages semantic correlations within the label hierarchy to enhance the performance of skin lesion diagnosis. 2) In the setting of $R^2 = 80\%$, it only requires coarse-level annotations (just identifying whether the lesion is melanocytic or non-melanocytic) for 80% of training data, and our LHT model results in an absolute performance gain of up to 8.6%. This

highlights the significant potential of our method to reduce the annotation cost for computer-aided diagnosis systems.

In Table X, we further compare the proposed LHT with state-of-the-art methods for skin lesion diagnosis on ISIC2018 dataset. The comparison methods include HD-CNN [24], HMCN [55], C-HMCNN [46], FGoN [27], and HRN [29], which are described in detail in Section IV-A2. The results show that our LHT model consistently outperforms all comparison methods over all five evaluation metrics. For example, we surpass the second best result by $79.9\% \rightarrow 82.4\%$ under $\mathcal{R} = 80\%$ and $73.7\% \rightarrow 75.8\%$ under $R^2 = 80\%$ on recall, highlighting that the proposed LHT method indeed helps to exploit the coarsely annotated hierarchical labels for skin lesion diagnosis. This reveals that our method can effectively leverage the training data annotated at different hierarchical levels to train a computer-aided diagnosis system. As such, the annotators may only need to provide partial labels with the highest confidence based on their own expertise, rather than annotating every sample into the finest-level label, which largely alleviates the conformation bias arising from noisy labels during training.

VI. CONCLUSION AND DISCUSSION

In this work we have proposed a probabilistic approach to address the hierarchical classification problem. Our key insight is to directly encode the correlation between two adjacent class hierarchies as conditional probabilities, such that the category labels across different hierarchies can be recursively predicted in a fine-to-coarse order. Beyond that, we have designed a confusion loss to further encourage the deep classifiers to learn the hierarchical correlation. We demonstrate the superiority of our method in hierarchical classification on a series of benchmark datasets and further shed light on the potential of the proposed method in computer-aided diagnosis.

Despite these promising results, our method has certain limitations. First, it relies on predefined hierarchical structures, which may not always be available in real-world applications. Second, the current framework focuses primarily on single-label classification, and its scalability to multi-label settings or to more general graph-structured hierarchies (e.g., cases where certain nodes have multiple parent categories) remains to be fully investigated. These limitations point to several promising directions for future research, including incorporating automatically discovered or partially observed hierarchies and extending the framework to multi-label and graph-structured scenarios.

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